

ECEN 743: Reinforcement Learning

Trust Region Methods

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References

- [SB, Chapter 13]
- [MMT, Chapter 12]
- [AJKS, Section 3]

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- We know that classical policy iteration algorithm gives monotone improvement guarantee
- However, due to the unknown MDP and large state space, greedy improvement of $Q_\pi(s, a)$ for each state-action pair is infeasible
- How do we ensure provable improvement guarantees for policy optimization algorithms?

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Lemma (Performance Difference Lemma)

Let $V_{\pi,\mu} = \mathbb{E}_{s \sim \mu}[V_{\pi}(s)]$. For any two policies, π and $\bar{\pi}$,

$$V_{\pi,\mu} - V_{\bar{\pi},\mu} = \frac{1}{(1-\gamma)} \mathbb{E}_{s \sim \rho_{\pi,\mu}(\cdot)} \mathbb{E}_{a \sim \pi(s,\cdot)} [A_{\bar{\pi}}(s,a)]$$

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$$V_{\pi}(s) - V_{\bar{\pi}}(s) = \mathbb{E}_{\tau \sim P_{\text{traj}, \pi}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] - V_{\bar{\pi}}(s)$$

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Now, averaging over the initial state s with a given initial state distribution μ , we get

$$\begin{aligned} V_{\pi, \mu} - V_{\bar{\pi}, \mu} &= \frac{1}{(1 - \gamma)} \mathbb{E}_{(s', a') \sim \rho_{\pi, \mu}} [A_{\bar{\pi}}(s', a')] \\ &= \frac{1}{(1 - \gamma)} \mathbb{E}_{s \sim \rho_{\pi, \mu}} \mathbb{E}_{a' \sim \pi(s, \cdot)} [A_{\bar{\pi}}(s, a)] \end{aligned}$$

Why PDL is Useful?

- PDL: $V_{\pi,\mu} - V_{\bar{\pi},\mu} = \frac{1}{(1-\gamma)} \mathbb{E}_{s \sim \rho_{\pi,\mu}} \mathbb{E}_{a \sim \pi(s,\cdot)} [A_{\bar{\pi}}(s, a)]$

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 - ▶ We know A_{π_k} and we can optimize over π ;
 - ▶ However, we don't know $\rho_{\pi,\mu}$ and evaluating that expectation requires sampling according to π

Trust Region Methods

- “Ideal” policy update step:

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- Will this “practical” policy update step give an improvement?

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- In CPI, we get $\|\pi_{k+1}(s, \cdot) - \pi_k(s, \cdot)\|_1 \leq 2\alpha$

Conservative Policy Iteration

- “Practical” policy update step:

$$\pi_{k+1} = \arg \max_{\pi \in \text{trust-region}(\pi_k)} \mathbb{E}_{s \sim \rho_{\pi_k, \mu}} \mathbb{E}_{a \sim \pi(s, \cdot)} [A_{\pi_k}(s, a)]$$

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Theorem (Monotone improvement)

Let $\bar{A}_k = \mathbb{E}_{s \sim \rho_{\pi_k, \mu}} \mathbb{E}_{a \sim \pi'(s, \cdot)} [A_{\pi_k}(s, a)]$. Then, $V_{\pi_{k+1}, \mu} - V_{\pi_k, \mu} \geq \frac{\alpha}{(1-\gamma)} \left(\bar{A}_k - \frac{2\alpha\gamma}{(1-\gamma)^2} \right)$.

Set $\alpha = \frac{\bar{A}_k(1-\gamma)^2}{4\gamma}$. Then, $V_{\pi_{k+1}, \mu} - V_{\pi_k, \mu} \geq \frac{\bar{A}_k^2(1-\gamma)}{8\gamma}$.

Auxiliary results

- We will state the following supporting result without proof (For proof, see [AJKS], Chapter 12)

Lemma

Suppose we have $\|\pi(s, \cdot) - \pi_k(s, \cdot)\|_1 \leq 2\alpha$ for all s . Then, we have

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- We will also use the following simple result

Lemma

Let z be a random variable and p, q be two distributions. Then,

$$|\mathbb{E}_{z \sim p}[f(z)] - \mathbb{E}_{z \sim q}[f(z)]| \leq \|p - q\|_1 \max_z |f(z)|$$

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Select α to get the maximum improvement. This will complete the proof.

□

TRPO and PPO

References

- Schulman, et al. “Trust region policy optimization”, *International Conference on Machine Learning (ICML)*, 2015.
- Schulman, et al. “Proximal policy optimization algorithms”, 2017.
- [AJKS], Section 3

Some Distance Metric

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- TRPO performs parameter update by solving the optimization problem

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- This reduces to NPG

$$\theta_{k+1} \leftarrow \theta_k + \sqrt{\frac{\alpha}{(\nabla V_{\pi_{\theta_k}})^{\top} F(\theta_k)^{-1} (\nabla V_{\pi_{\theta_k}})}} F(\theta_k)^{-1} (\nabla V_{\pi_{\theta_k}})$$

Importance Sampling

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- One obvious approach is: generate n i.i.d. samples $(z_i)_{i=1}^n$ according to p . Then,
$$\mathbb{E}_{z \sim p}[f(z)] \approx \frac{1}{n} \sum_{i=1}^n f(z_i)$$

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$$\mathbb{E}_{z \sim p}[f(z)] \approx \frac{1}{n} \sum_{i=1}^n f(z_i)$$
- Suppose we can only generate i.i.d. samples according q . How do we get the estimate for $\mathbb{E}_{z \sim p}[f(z)]$?

Importance Sampling

- We have

$$\mathbb{E}_{z \sim p}[f(z)] = \mathbb{E}_{z \sim q}\left[\frac{p(z)}{q(z)} f(z)\right] \approx \frac{1}{n} \sum_{i=1}^n \frac{p(z_i)}{q(z_i)} f(z_i), \quad \text{where, } z_i \sim q$$

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$$\begin{aligned} \text{Var}\left(\frac{p(z)}{q(z)} f(z)\right) &= \mathbb{E}_{z \sim q}\left[\left(\frac{p(z)}{q(z)} f(z)\right)^2\right] - \left(\mathbb{E}_{z \sim q}\left[\frac{p(z)}{q(z)} f(z)\right]\right)^2 \\ &= \mathbb{E}_{z \sim p}\left[\frac{p(z)}{q(z)} f^2(z)\right] - (\mathbb{E}_{z \sim p}[f(z)])^2 \end{aligned}$$

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- What is the variance of the importance sampling based estimate?

$$\begin{aligned} \text{Var}\left(\frac{p(z)}{q(z)} f(z)\right) &= \mathbb{E}_{z \sim q}\left[\left(\frac{p(z)}{q(z)} f(z)\right)^2\right] - \left(\mathbb{E}_{z \sim q}\left[\frac{p(z)}{q(z)} f(z)\right]\right)^2 \\ &= \mathbb{E}_{z \sim p}\left[\frac{p(z)}{q(z)} f^2(z)\right] - (\mathbb{E}_{z \sim p}[f(z)])^2 \end{aligned}$$

- Importance sampling weight, $p(z)/q(z)$, can be very large for some z . Then the variance can be very large.

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Reformulation of the TRPO Objective

- Restating with TRPO Objective with a minor change in the trust region (average KL to max KL):

$$\begin{aligned} \max_{\theta} \quad & \mathbb{E}_{s \sim \rho_{\pi_{\theta_k}, \mu}} \mathbb{E}_{a \sim \pi_{\theta}(s, \cdot)} [A_{\pi_{\theta_k}}(s, a)] \\ \text{s.t.} \quad & D_{TV}^{\max}(\pi_{\theta_k}, \pi_{\theta}) \leq \alpha \end{aligned}$$

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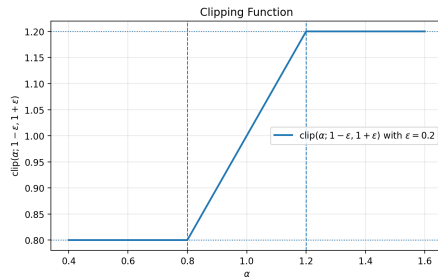
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- How do we ensure the trust region constraint?

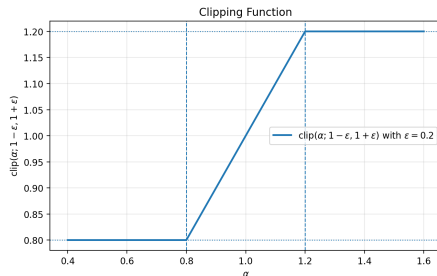
Clipping

$$\text{clip}(\alpha; 1 - \epsilon, 1 + \epsilon) = \begin{cases} 1 - \epsilon, & \alpha \leq 1 - \epsilon, \\ 1 + \epsilon, & \alpha \geq 1 + \epsilon, \\ \alpha, & \text{otherwise.} \end{cases}$$



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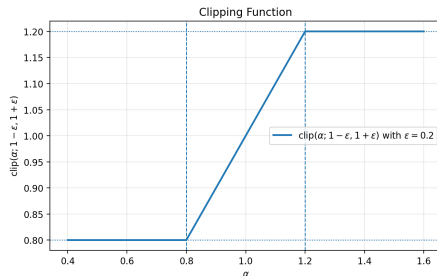


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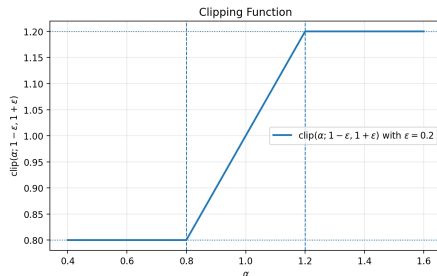


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- This can be solved by sgd

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 - ▶ Suppose $\alpha_t(\theta) \gg 1 - \epsilon$, the new policy has a high probability to this bad action
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Better Clipping

- The sign of A_t should determine the desired direction of change:
 - ▶ If $A_t > 0$, then the sampled action is better than expected, so increasing its probability is beneficial
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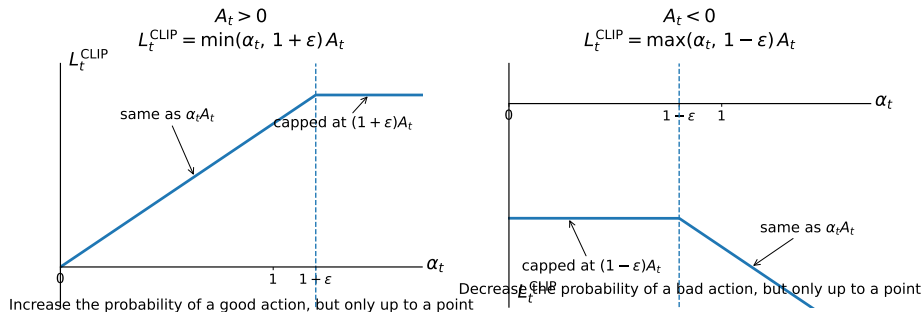
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- This is pessimistic surrogate:
 - ▶ No extra credit for a large policy change in the direction that appears beneficial
 - ▶ Fully penalizes large changes in the harmful direction

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